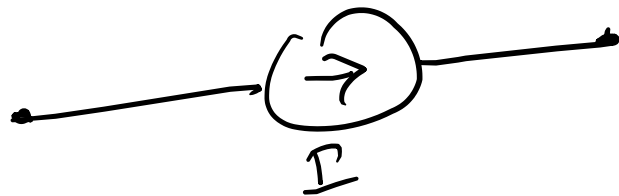
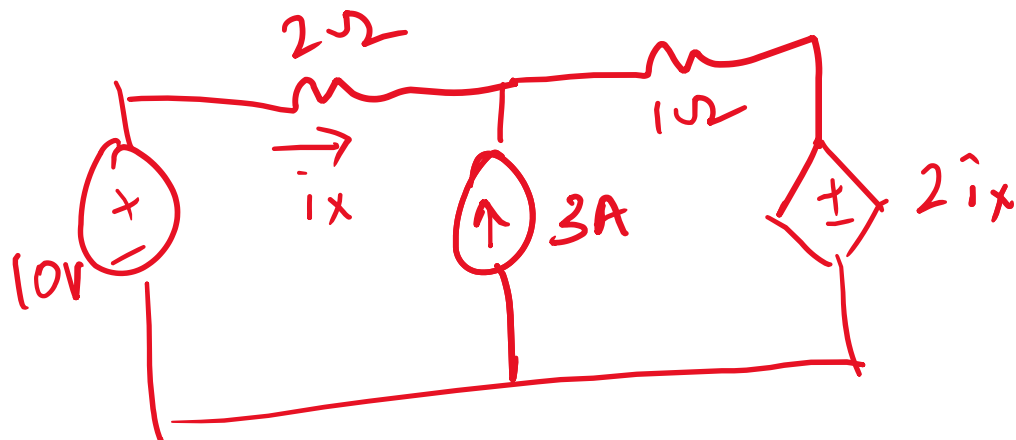
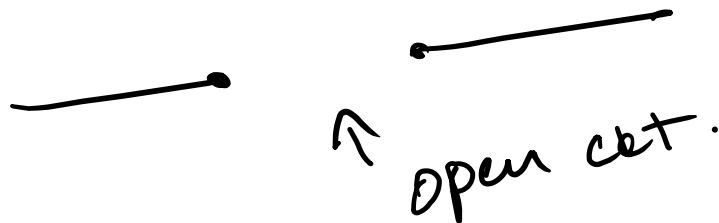


Superposition

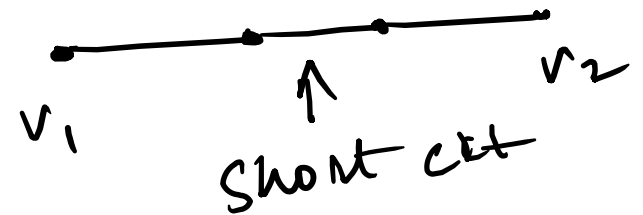


$$I = 0$$



$$V = 0$$

$$v_1 = v_2$$



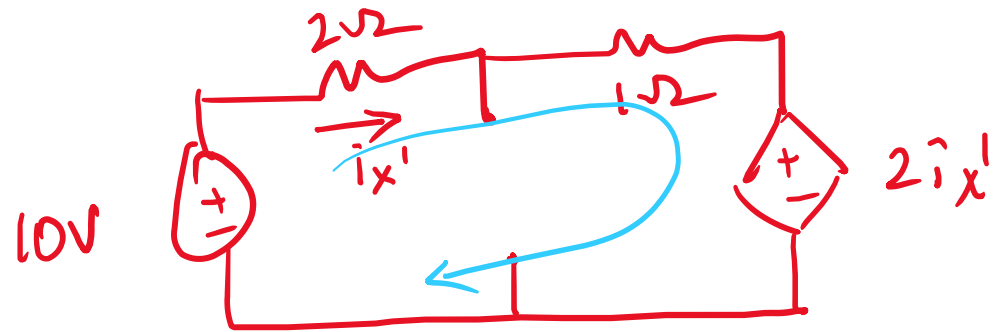
Find i_x using superposition.

No. of ind sources: 2
(10V, 3A)

$$i_x = i_x' + i_x''$$

(10V) (3A)

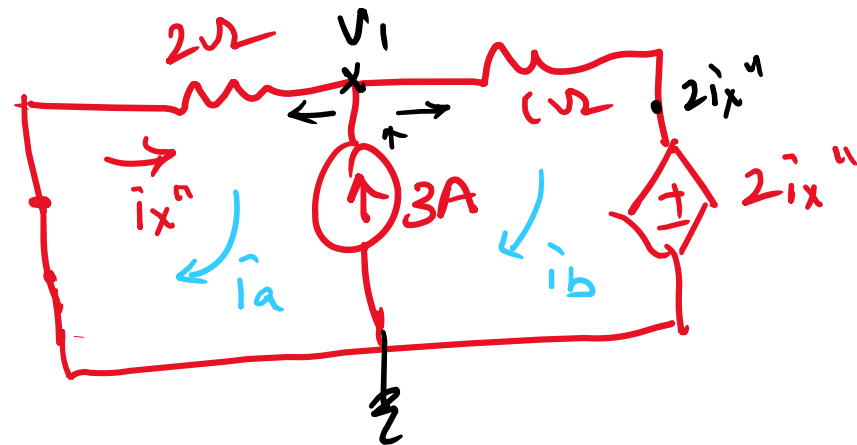
10V ON, 3A OFF.



$$10 - 2\hat{i}_{x'} - 1\hat{i}_{x'} - 2\hat{i}_{x'} = 0$$

$$\hat{i}_{x'} = 2A$$

3A ON, 10V OFF



$$\hat{i}_{x''} = -\frac{V_1}{2}$$

$$\hat{i}_{x''} = \hat{i}_a$$

At V_1 :

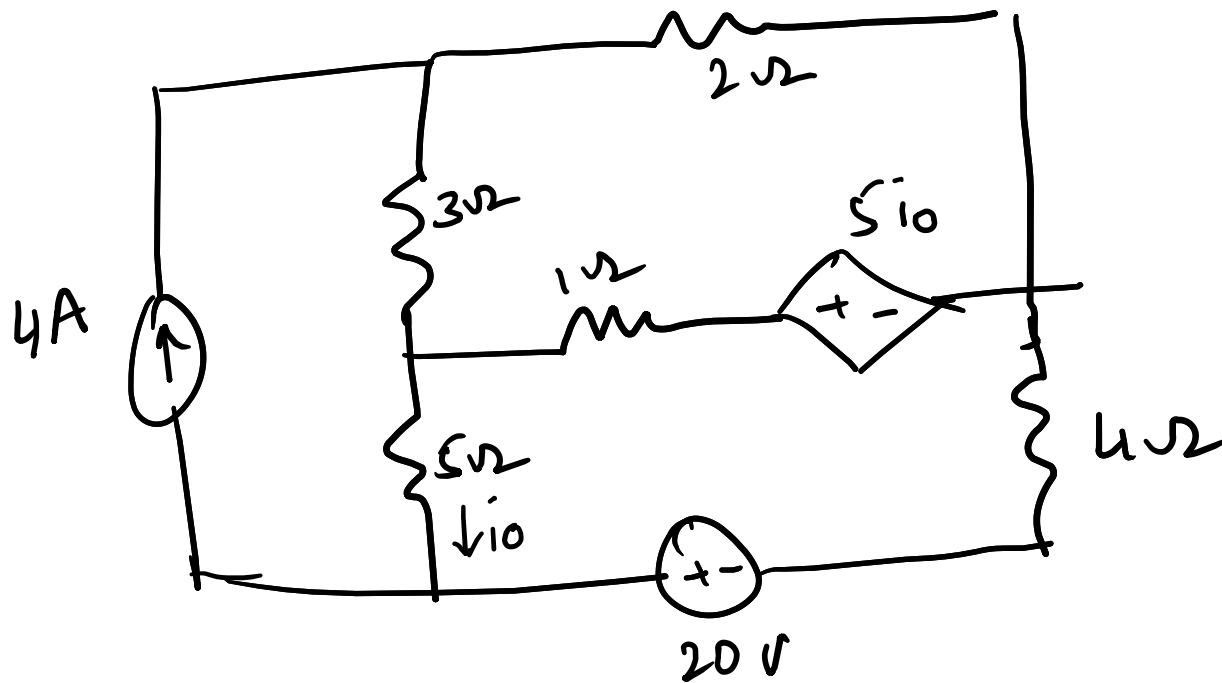
$$3 = \frac{V_1 - 0}{2} + \frac{V_1 - 2\hat{i}_{x''}}{1}$$

$$\hat{i}_{x''} = \underline{\underline{-0.6A}}$$

$$\hat{i}_b - \hat{i}_a = 3A$$

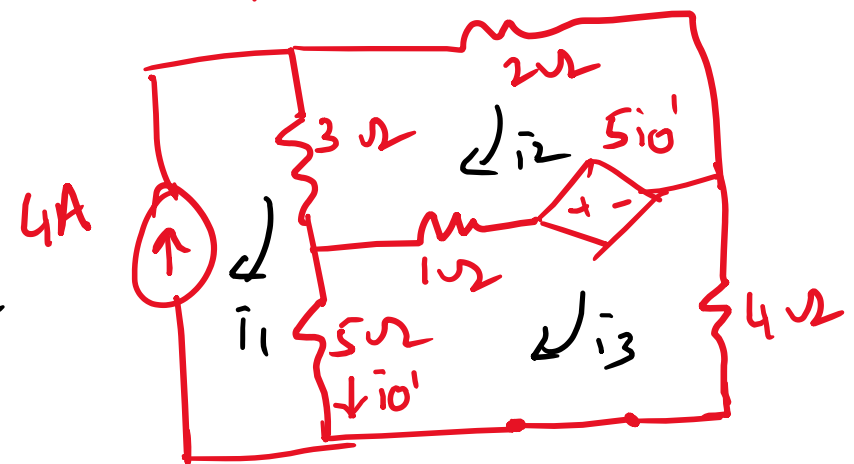
$$-2(\hat{i}_a) - 1(\hat{i}_b) - 2\hat{i}_{x''} = 0$$

$$\hat{i}_x = \hat{i}_{x'} + \hat{i}_{x''} = \underline{\underline{1.4A}}$$



Find i_o using superposition.

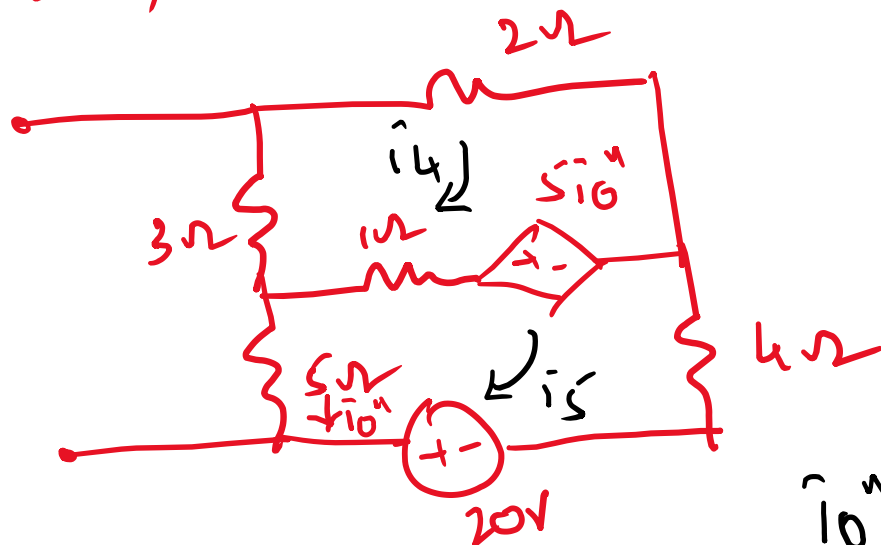
4A ON, 20V OFF.



$$i_1 = 4A$$

$$i_o' = i_1 - i_3$$

4A OFF, 20V ON



$$i_o'' = -i_5$$

Mesh 2:

$$5i_o' - 1(i_2 - i_3) - 3(i_2 - i_1) - 2i_2 = 0$$

Mesh 3:

$$-5i_o' - 4i_3 - 5(i_3 - i_1) - 1(i_3 - i_2) = 0$$

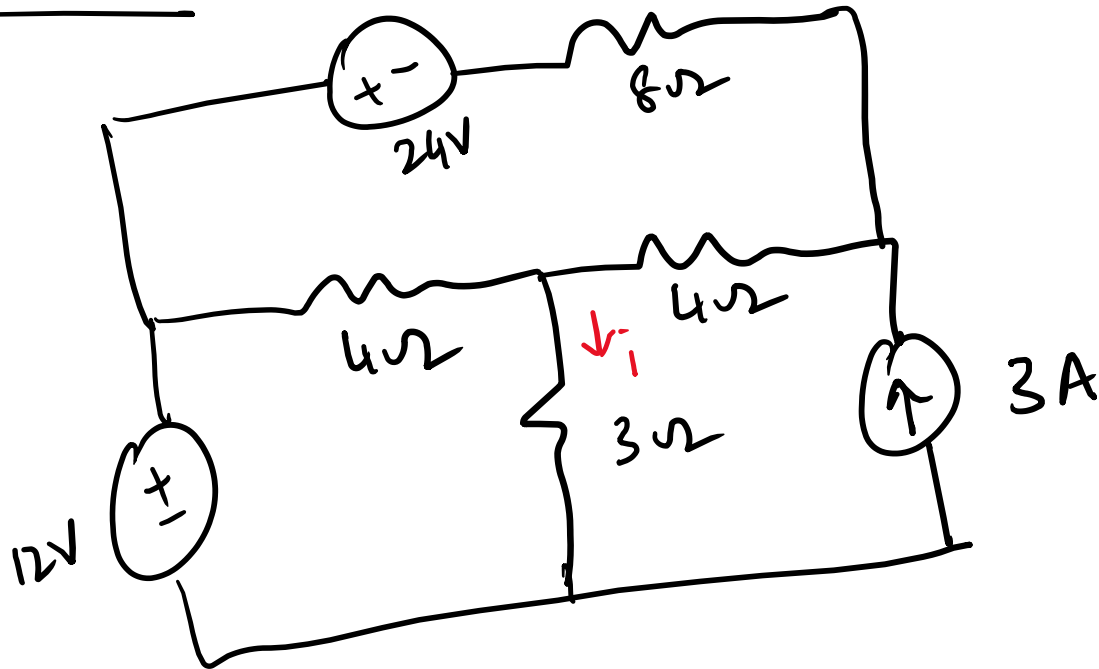
$$i_o' = \underline{\underline{3.05A}}$$

Mesh 4: $5\hat{i}_0'' - 1(\hat{i}_4 - \hat{i}_5) - 3(\hat{i}_4) - 2(\hat{i}_4) = 0$

Mesh 5: $20 - 5\hat{i}_5 - 1(\hat{i}_5 - \hat{i}_4) - 5\hat{i}_0'' - 4\hat{i}_5 = 0$

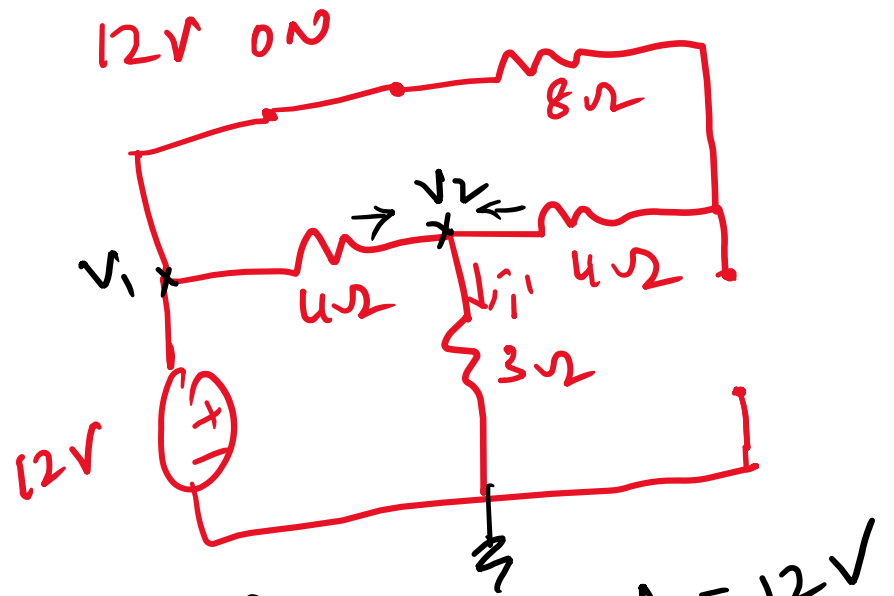
$\hat{i}_0'' = -3.5A$

$\hat{i}_0 = \hat{i}_0' + \hat{i}_0'' = \underline{\underline{-0.47A}}$



$\hat{i}' = \frac{v_2}{3}$

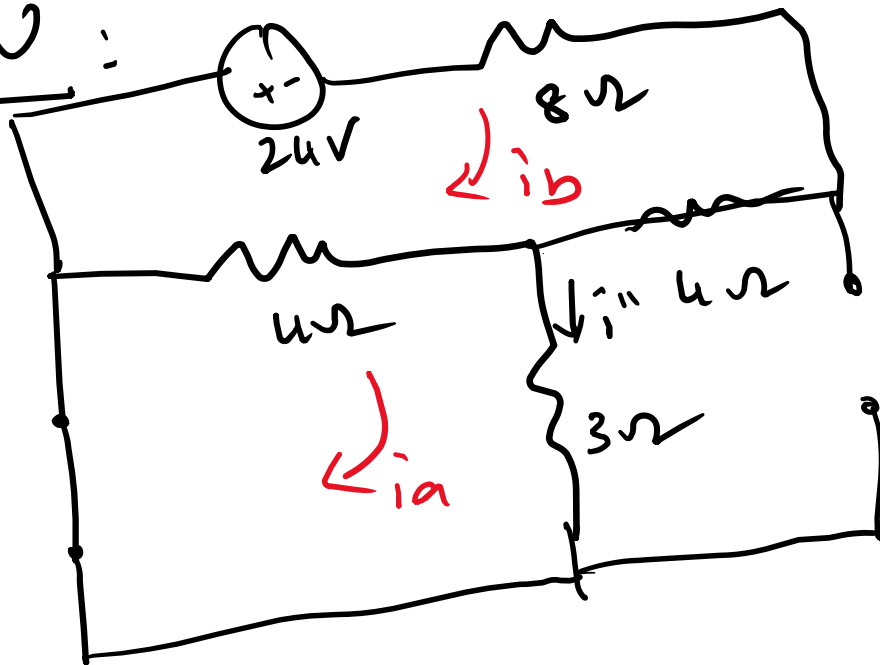
Find i using superposition



$\frac{v_1 - v_2}{4} + \frac{v_1 - v_2}{8 + 4} = \frac{v_2}{3}$

$v_1 = 12V$

24V ON :

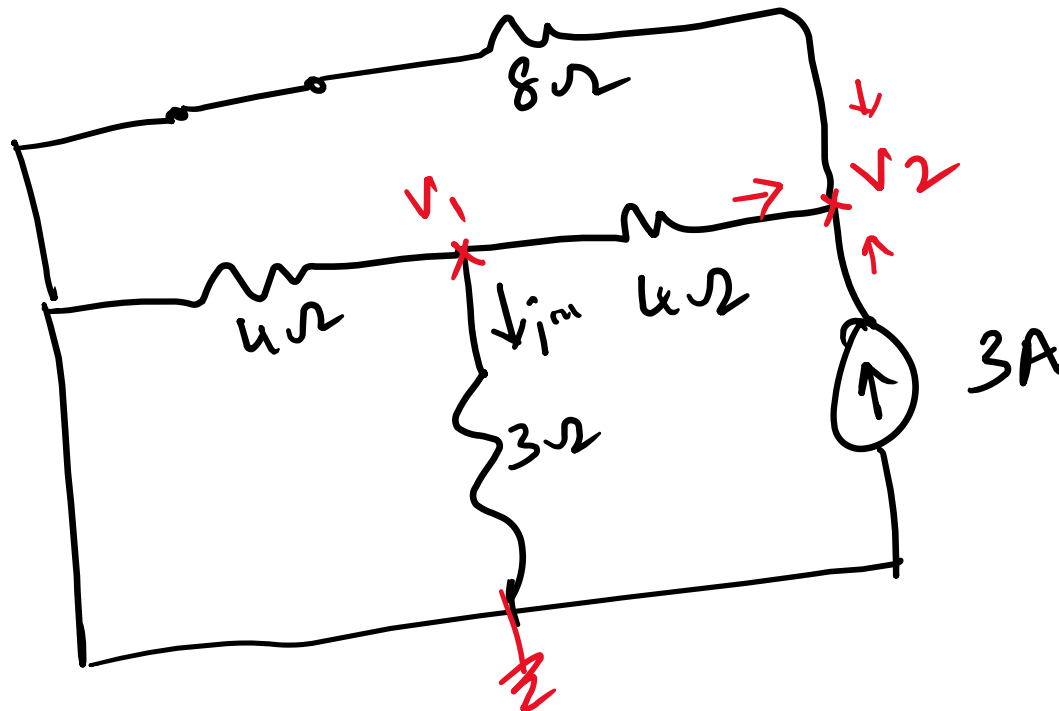


$$i'' = i_a$$

$$-4(i_a - i_b) - 3i_a = 0$$

$$-24 - 8i_b - 4i_b - 4(i_b - i_a) = 0$$

3A ON :



$$i'' = \frac{V_1}{3}$$

At V_1

$$\frac{V_1}{3} + \frac{V_1 - V_2}{4} + \frac{V_1}{4} = 0$$

At V_2 :

$$3 + \frac{V_1 - V_2}{4} + \frac{0 - V_2}{8} = 0$$